Whose Contribution Counts?

AI Mediation and Collective Authorship in Digital Improvisation

Roman Jahandideh

Exploratory proposal for discussion with Dr. Rebecca Caines
Graduate Program in Communication & Culture, York University
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Working proposal prepared for discussion with Dr. Rebecca Caines. The research setting and project partnerships will be developed in consultation with the prospective supervisor.

1 Problem and cultural stakes

Generative models now sit inside tools that people use to make things together, and in these tools a model does more than produce content: it *reinterprets* what people contribute. Someone types a phrase, and a moment later it has become a sound or a moving form in a shared scene. The group then has to decide what that transformation means, and whether it still belongs to the person who offered it.

The stakes reach beyond any single workshop. When AI translates creative contributions into shared audiovisual material, its outputs may become more visible or influential than the contributions that prompted them. I treat this as a possibility to investigate, not an established effect. If it does happen, it raises a cultural question about whose interpretations gain authority, whose contributions remain recognisable, and how disagreement is negotiated in collective creative practice.

Improvisation is a good place to ask this question, because improvisers already have ways of listening to, building on and refusing one another's offers. Improvisation has also been argued to be a research method in its own right, one grounded in risk, active listening and collaborative response (Caines, 2020). A machine that transforms offers enters these practices, and the people involved have to work out where it fits.

2 Research gap

Existing research examines human–AI co-creation, collective decision-making and the attribution of creative credit. This project investigates their intersection in small-group digital improvisation: how a contribution becomes recognisable as someone's work, how its transformation can be challenged, and how authority over the resulting artifact is negotiated over repeated sessions. Three studies come closest to this question, and each leaves a different part of it open.

Xu et al. (2026) observed six groups of three university students designing a cultural educational game around one shared ChatGPT interface, in a single 45-minute session, and interviewed nine further students. Shared

AI output became an anchor for attention, a shared memory and a negotiable part of the groups' decisions. The authors note that their analysis emphasised information flow over social aspects such as trust and accountability. I take their question into creative work in which a specific person's material is transformed, and I follow authorship and accountability across repeated sessions.

Trajkova et al. (2024) ran focus groups with 24 dance students about partnered movement improvisation, in order to design an AI dance partner. In that work the AI is a *partner*. In mine it is a *mediator*, reinterpreting one person's contribution for a whole group.

Epstein et al. (2020) showed, through online vignette experiments, that describing an AI as an agent rather than a tool shifts the credit and responsibility that observers assign to the people behind an AI artwork. Their participants judged a finished work from the outside. I ask how credit and responsibility are claimed and contested *in situ*, by the people making the work.

Questions of contestation point the same way. Contestability has mainly been developed for high-stakes algorithmic decisions such as hiring or sentencing (Lyons et al., 2021). Much less has been said about what it means to challenge a machine's *creative* reading of one's contribution, in front of collaborators.

3 Research questions

Central question. In collaborative digital improvisation, how do participants, facilitators and a generative-AI interface shape whether a person's contribution is recognised, transformed, taken up, contested and credited in the shared work?

RQ1. How do participants interpret differences between what they offered and how it was transformed?

RQ2. How do groups and facilitators negotiate acceptance, revision, disagreement and refusal, and who comes to have the standing to decide?

RQ3. How do these negotiations shape felt ownership, public attribution and creative responsibility for the resulting work?

Two terms need defining. On **participation**: I focus on the trajectory of identifiable creative contributions, while also attending to listening, facilitation, silence and refusal as forms of participation. Recognition will be examined through participants' interpretations and interactions, rather than inferred solely from incorporation into the artifact. I do not assume that group decision-making is fair, that individual control is enough, or that facilitation does harm. A majority can override someone; a facilitator can help a quieter participant take part.

By **authorship** I mean three related things: *felt ownership*, whether a person experiences part of the work as theirs; *public attribution*, how the work is credited; and *creative responsibility*, who answers for what the work does. Legal ownership and copyright are outside the scope of this study.

4 Theoretical framework

My primary framework is Lucy Suchman's study of human–machine reconfiguration (Suchman, 2007), which examines how agencies are figured at the interface and how humans and machines come to be treated as similar or different in practice. Read this way, the question of whether a transformed element "belongs" to its contributor, to the group or to the system is not settled in advance. People work it out as they interact. This gives me three questions to bring to the material:

- how participants attribute agency and authorship in talk and action ("the AI did that", "that one is mine");
- how interface rules, model defaults, facilitation and my own design decisions make some actions easy and others difficult;
- who is held responsible when a transformation misrepresents someone.

Improvisation studies adds a limited, complementary lens: attention to listening and response, and to how an offer is taken up, ignored or refused (Caines, 2020). These concepts tell me where to look. They do not decide what I will find.

5 Relationship to my master's research

At Simon Fraser University, my master's research has asked how interfaces make a person's state perceptible to others. In a cooperative Unity game, I rendered remote teammates' heart rate and heart-rate variability as label-free ambient icons; in a second project, I built a Unity environment that responds to the player's heart rate. Both manuscripts were submitted to ACM venues and received revision requests. Neither has been accepted (Jahandideh, 2026b,a).

The teammate study left me with a question I could not answer inside it. Participants tied their willingness to use the cue to control over it: they asked for opt-in sharing, an immediate off switch and separate permissions for the icon and the numbers (Jahandideh, 2026b). Looking back as the designer, I also noticed that the classification behind the label-free display still sorted people into evaluative categories they never saw. Both observations lead to the question this PhD takes up: who controls how a shared system interprets people?

The PhD does not continue physiological sensing. It moves from involuntary bodily signals to deliberate creative contributions, and from consent to being made visible to the negotiation of being transformed. My training fits this shift. In architecture and landscape architecture I learned to design how people perceive and move through space; in HCI I have built and evaluated networked interactive systems myself. That combination prepares me to design a shared environment as a creative work, and then to study how people interpret it.

6 Research-creation methodology

The project follows the program's research-creation option, which combines a creative work with a written dissertation of 150–200 pages that develops the argument (Joint Graduate Program in Communication & Culture, 2021). Of the modes Chapman and Sawchuk (2012) describe, it is closest to "research-from-creation": making the work generates material for understanding how participants engage with it.

The creative component will comprise the shared audiovisual environment and the improvised works produced through it. Developing and revising its expressive possibilities, interaction rules and workshop scores will form part of the inquiry into how creative authority is distributed. What participants make are short collective audiovisual improvisations: projected, spatial scenes of light, form and sound that they build and perform together in real time, and then talk about.

A brief, illustrative session shows what this looks like. Four improvisers receive a score: build the weather of a city none of them has visited. One types "the bus that never comes." The system proposes a slow amber pulse at the edge of the space with a low drone, and displays its generated interpretation, "waiting as a distant repeating light." The contributor accepts it. Later, another participant wants the pulse brighter. The contributor declines, and the facilitator asks the group how to proceed. Moments like these, of acceptance, challenge and resolution, are what the study examines.

The work unfolds in four phases:

1. *Scoping interviews* with improvisers and facilitators about how offers, refusal and credit work in their practice, with and without technology. What I learn shapes the scores and the system's rules.
2. *Co-design* sessions with a few practitioners to choose the input form, set limits on the transformations, design controls for refusal and credit, and test accessibility provisions.
3. *Repeated workshops* with two or three small groups, each meeting several times so that shared norms can form. The number of groups and sessions will depend on access and on the material that emerges.
4. *Reflection interviews* in which participants review video episodes from their own sessions.

Each group works under three arrangements, in an order that varies across sessions:

Arrangement	Who proposes the transformation	Who approves it
Contributor-led	AI system	The contributor
Group-led	AI system	The group, by a procedure it agrees in its first session
Human-mediated	Another participant in a rotating "interpreter" role, using the same asset library	The contributor (same rule as contributor-led)

The contributor-led and group-led arrangements explore different distributions of approval authority. The human-mediated arrangement, which uses comparable assets and the same approval rule as the contributor-led one, offers an additional interpretive contrast. It also changes timing, social relationships and expectations.

These workshops will therefore support situated comparison rather than isolate a causal effect of AI, and a study of this size does not generalise statistically.

7 The system and the place of AI

A participant submits a short text phrase; sketches may be added later if co-design supports them. An existing language model returns one *proposal*, a choice of elements and settings from a curated library of forms, light, colour, particle "weather", sound and text, together with a one-line *generated interpretation* of the phrase that everyone can see. That line is part of the model's output. It does not explain why the model produced the transformation, which it cannot establish. Depending on the arrangement, contributors or groups can inspect, edit, accept, reject, request another proposal or undo. A manual path to the same library is always open, so no one needs AI in order to contribute.

Generative AI is included because machine-mediated reinterpretation is the object of study. The project does not presume that AI improves improvisation; it examines how participants respond to its transformations and negotiate their place in the shared work. The model does not decide whether anything enters the work, who is credited or what is removed, and it makes no inferences about participants' emotions or behaviour.

Feasibility. Unity applies only a fixed set of permitted actions with bounded settings. Model output is checked against that set and never run as code; malformed output or a service failure simply produces no proposal. Each proposal and decision is recorded and can be undone. I will first test a locally run open-weight model, which avoids third-party processing, and otherwise use a hosted service under institutionally approved terms, sending only the phrase. I will not train or fine-tune models: this is a research-creation project built on existing tools, not an AI engineering dissertation.

8 Case-study context and access

Possible primary context, subject to Dr. Caines's agreement: workshops with adult improvising artists and creative-technology practitioners in Toronto, connected to her SSHRC Insight project *Interfacing, Playtesting, Resetting: Reframing Technology Use Through Improvised Creative Technologies* (Goodfellow Craig, 2026). Only the project's title and funding are public. I do not assume that it includes this study, these participants or any AI component; whether a student study could sit alongside it is for her to judge.

Alternative: open calls to Toronto improvisation, theatre and sound communities and to AMPD students, excluding anyone I teach, with an arts organisation as host only if it agrees. Earlier projects such as multiPLAY are precedents for digitally mediated improvisation, not recruitment sources. I will not work with disability-focused or Indigenous communities without established partnerships, community governance and relevant expertise.

9 Analysis

I will construct multimodal accounts of selected episodes, linking participants' words and actions with screen events and system records. An episode runs from a contribution, through its transformation and the group's response, to the next creative action. Analysis will compare accepted transformations, contested transformations, refusals, and episodes in which participants disagree about authorship. Later interviews will add participants' own interpretations, including accounts that challenge my initial reading, and comparisons across sessions will show how these practices change as groups establish shared norms.

For the episodes I adapt the video microethnography that McArthur (2015) used to study players' interactions with interfaces. Interviews will be analysed through reflexive thematic analysis (Braun and Clarke, 2019). System records show what happened in the software; they are not a "true" account of intention against which participants are judged. When records, observation and participants' accounts diverge, the divergence is itself a finding.

10 Ethics, consent and accessibility

Consent will be specific, ongoing and revisable (Im et al., 2021; Strengers et al., 2021). Participants consent separately to recording, to AI processing of their contributions, and to public dissemination of the works.

Withdrawal needs particular care in collective work. Provenance will support identifying and, where feasible, removing attributable contributions. Participants will agree on withdrawal procedures and deadlines before the workshops, including the limits of removing contributions from jointly developed or publicly disseminated works. Credit for public works will be agreed with participants, and their time will be paid. Data will be minimised and pseudonymised, any third-party processing disclosed, and retention periods stated.

Full participation without AI is a design commitment that the workshops will test. Planned provisions are a reduced-motion setting, readable and resizable text, visual or textual alternatives to sound-only information, controls that a facilitator can operate on a participant's behalf, and the manual path.

Image-generation models can reproduce stereotypes even when prompts avoid demographic terms (Bianchi et al., 2023), so I will discuss model defaults openly with participants. Participation alone does not make a project equitable (Birhane et al., 2022). Finally, as designer, researcher and possibly facilitator, I shape the situation I study. I will keep a design journal and, where possible, work with a co-facilitator.

11 Intended contributions

The intended scholarly contribution is an account of how creative authority is negotiated across contributors, facilitators, interface rules and AI-generated transformations. The study will examine when technical attri-

bution supports recognition, and when it fails to resolve disagreement about authorship. The creative system and workshop documentation will make these tensions available for analysis and further practice.

These are aims, not findings. The workshops may show that AI is unnecessary in some arrangements. If so, the contribution would lie in explaining when, how and why it becomes unnecessary, and what that reveals about collective creativity.

12 Provisional chapter structure

1. Introduction.
2. Co-creation, credit and contestation: literature.
3. Framework: human–machine reconfiguration and improvised response.
4. Methodology, research-creation and reflexivity.
5. The environment as a creative work and a configured situation.
6. Recognition and transformation.
7. Disagreement, refusal and facilitation.
8. Ownership, attribution and responsibility.
9. Conclusion.

13 Timeline (full-time, typical program schedule)

Year 1 (2027–28)	Courses (CMCT 7000, 7200, 7005 and electives); literature review; ethics preparation; technical tests with a local model.
Year 2 (2028–29)	Comprehensive examinations; formal dissertation proposal (maximum 3,500 words) with the supervisory committee (Joint Graduate Program in Communication & Culture, 2016); ethics approval; scoping interviews.
Year 3 (2029–30)	Co-design; repeated workshops; reflection interviews; ongoing analysis.
Year 4 (2030–31)	Analysis; documentation or public sharing of the creative work; writing and defence. Year 5 is held in reserve.

14 Program and supervisor fit

The project belongs to the program's Technology in Practice area, with Media & Culture as a minor. It approaches AI-mediated creative work as a cultural practice of participation, recognition and authorship, rather than as a technical problem.

Dr. Caines's research on improvisation as a research methodology (Caines, 2020) and on socially responsive creative technologies (Caines and Watson, 2023) speaks directly to the project's setting and methods. Her recent research-creation study of generative AI in interdisciplinary classrooms attends to ethics, access, col-

laboration and improvisatory skills (Caines, 2026). My project carries that concern from learning with these tools to contribution-level recognition and contested authorship in repeated group improvisation.

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